THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

NMTC – 2026

RAMANUJAN CONTEST – INTER FINAL

Instructions:

1. Answer all questions. Each question carries 10 marks.
2. Elegant and innovative solutions will get extra marks.
3. Diagram and justification should be given wherever necessary.
4. Before start answering, fill in the FACE SLIP completely.
5. Your rough work should be in the answer sheet itself.
6. Maximum time allowed is THREE hours.

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1. Find all positive inetgers m, n such that $(m^2 - n^2)^2 = 23n + 8$.

2. One hundred boys and one hundred girls go to Navarathri Garba dance. Knowing that each boy dances with a girl at most once and that there are 1050 couples danced, prove that there are two boys and two girls who danced with both the boys.

3. Positive real number sequences a_n and b_n are defined by

$$a_{n+1} = a_n - \frac{1}{1 + \sum_{i=1}^{n} \frac{1}{a_i}}$$

$$b_{n+1} = b_n + \frac{1}{1 + \sum_{i=1}^{n} \frac{1}{b_i}}$$

   If $a_{100} = b_{99}$, show that $a_{100} + b_{100} > a_{101} + b_{101}$.

4. Let $\mathbb{N}_+$ be the set of positive integers and the function $f : \mathbb{N}_+ \to \mathbb{N}_+$ satisfy the following conditions:

   (a) $f(n) < f(n+1)$ for all $n \in \mathbb{N}_+$;
   (b) $f(f(f(f(n)))) = 5n$ for all $n \in \mathbb{N}_+$.

   Find the value of $f(2000)$.

5. Two circles S_1 and S_2 with centers O_1 and O_2 respectively intersect at X and Y. AB is an external common tangent to the circles with A and B lying on S_1 and S_2 respectively. Tangents to S_1 and S_2 at X intersect the line of centers O_1O_2 at L and K respectively. Show that XY, AK and BL are concurrent.

6. Let p and q be primes such that $p^2 + pq + q^2$ is a perfect square. Show that $p^2 - pq + q^2$ is a prime.

7. Prove that any set of 55 integers from the set $\{1, 2, \ldots, 100\}$ will contain a pair of numbers that differ by 9, a pair that differ by 10, a pair that differ by 12, and a pair that differ by 13. Will such a set of 55 integers also contain a pair of numbers that differ by 11?

8. Find all polynomials $P(x) = x^3 + ax^2 + bx + c$ where a, b, c are integers such that the following hold:

   (a) $x - 1$ divides $P(x)$;
   (b) $P(5) = P(-5)$;
   (c) $P(x) = 0$ has a root between 2 and 3.